Subwavelength Nano-imaging with a Microbottle Lens

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2022 Asia Communications and Photonics Conference (ACP) | 978-1-6654-8155-7/22/$31.00 ©2022 IEEE | DOI: 10.1109/ACP55869.2022.10088533

***Abstract*—A dielectric microbottle lens constituted by circular glass microcylinder and parabolic-shaped polymer covering layer is demonstrated here. The microbottle lens is fabricated by transferring, wetting and photocuring of UV-curable polymer on a tapered optical microfiber. By using such microbottle lens, periodic grating structures with subwavelength pitch of 360 nm can be imaged in two different imaging modes. When the pitch of the grating structures decreases to 260 nm, clear imaging of the grating structures can only be achieved in the case that the grating stripes are parallel to the microfiber axis of the microbottle lens. The results are analyzed and discussed by finite element simulation.**

Keywords-component; microbottle lens; subwavelength nano-imaging; grating stripes; finite element simulation

I. INTRODUCTION

Since the discovery of microsphere nanoscopy by Wang *et al.* in 2011, nano-imaging technology based on microparticle lenses has been developed for more than a decade [1]. Microparticle lens-assisted nano-imaging technology has received extensive attention in academic and production fields due to the advantages of label-free, no phototoxicity, wide-field imaging, low-cost setup, and fast imaging speed [2-6]. A series of microparticle lens-assisted nano-imaging systems including endoscopy [7], spider silk [8], spherical cap [9], cascaded microsphere lenses [10], and single-cell biomagnifier [11] were successfully applied to image various samples of interest, such as virus [12], biomolecules and subcellular organelles [13], sub-surface nanostructures [9, 14], semiconductor chips [15], plasmonic interaction [16], phase-shifting profilometry [17], nanomaterials [18], surface metrology [19], visualization of nanomanipulation [20], as well as used for observation of Brown motion[21], metrology of semiconductor devices [22], etc.

The nano-imaging capability of the microparticle lens is related to the curvature of particle's circular cross-section. The spherical microparticle lens is an isotropic imaging device, while the cylindrical microparticle lens has the imaging ability only in the circular cross-section direction [8]. In a sense, the microcylinder lens is formed by the infinite stretching of the microsphere lens along a certain straight direction. In order to find out whether there is attractive physical phenomenon between these two forms, we fabricate a composite microbottle lens made of polymer and silica, by transferring, wetting and photocuring of UV-curable polymer on a tapered optical fiber. In the direction along the microfiber axis and the direction perpendicular to the microfiber axis, the microbottle lens has the cross section of two different contour profiles. When the microfiber axis and grating stripes are arranged perpendicularly or on parallel,

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this microbottle lens can achieve two different imaging effects for grating stripes with subwavelength pitches. The imaging effect is analyzed and discussed by numerical simulation based on finite element method (FEM).

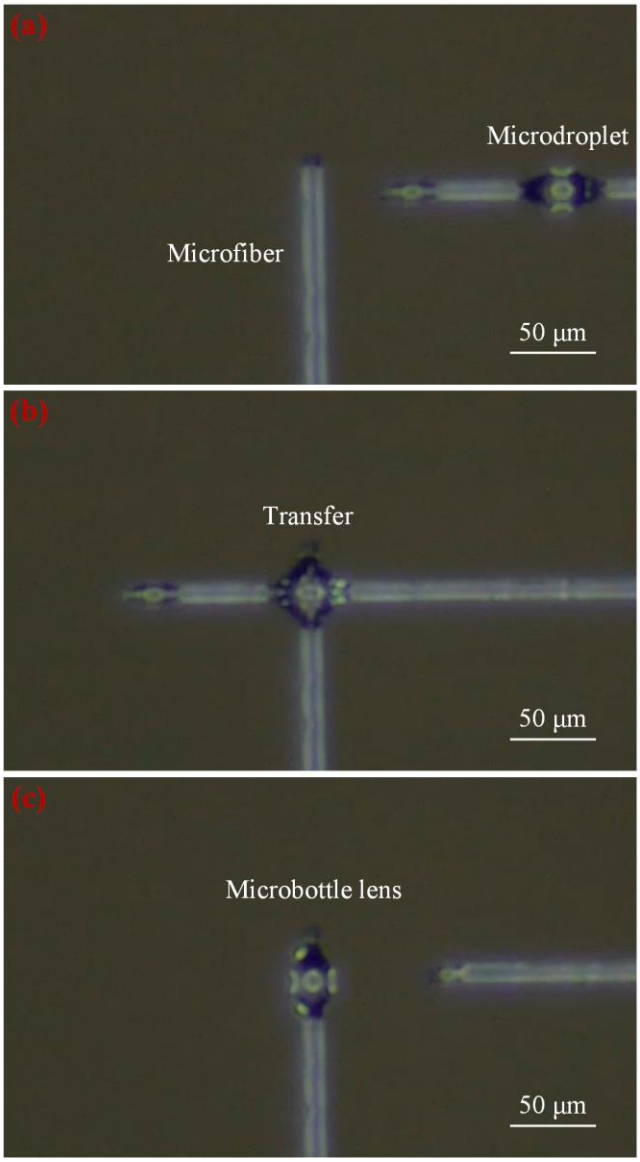

Figure 1. Experimental process of fabricating UV-curable adhesive microbottle lens.

II. Fabrication of Microbottle Lens

In our previous studies, it has been demonstrated that UV-curable adhesive can wet silica fiber and form bottle-shaped microresonators [23]. However, by dispersing a large adhesive droplet over a long region on a tapered microfiber brings difficulties to control the number and interspacing of the microresonators. If only one microbottle is used as the lens for imaging, it is affected or restricted by the adjacent microbottle lens. Therefore, here we fabricate only one microbottle lens near the tip of each microfiber for imaging. The experimental process for the fabrication of microbottle lens with Norland Optical Adhesive 61 (NOA 61) and silica fiber is shown in Fig. 1. First, the microdroplet attached to the microfiber (transverse direction) is transferred to the tip of the microfiber positioned along the longitudinal direction using a nano translation stage. Second, the transferred microdroplet forms a microbottle lens on the microfiber under the action of solid-liquid interfacial tension. Finally, the liquid microbottle lens is cured by the irradiation of a UV curing lamp.

Figure 2(a) shows the three-dimensional (3D) schematic diagram of the microbottle lens. The dimensions of the microbottle lens can be described by three parameters: the neck-to-neck distance L_b, the microbottle diameter D_b, and the microfiber diameter D_f. Figure 2(b) is the SEM image of the composite microbottle lens composed of UV-curable polymer (NOA 61) and microfiber (SiO_2) produced in our experiment. The values of these three parameters are L_b = 75 μm, D_b = 37 μm, and D_f = 19.6 μm, respectively. A parabolic profile of $D(z) = D_b\,[1\text{-}0.5(\Delta k{\cdot}z)^2]$, $|z| \le L_b/2$ and $D(z) = D_f$, $|z| > L_b/2$ with curvature Δk = 0.0259 μm^{-1} [the green dashed curve in Fig. 2(b)] can be used to fit the sectional shape of the microbottle lens along the axis of the microfiber. The shape and size of the composite microbottle lens are determined by several factors, such as the size and smoothness of the microfiber, the amount of NOA 61 to generate the structure, and the solid-liquid interfacial tension between the polymer and the microfiber.

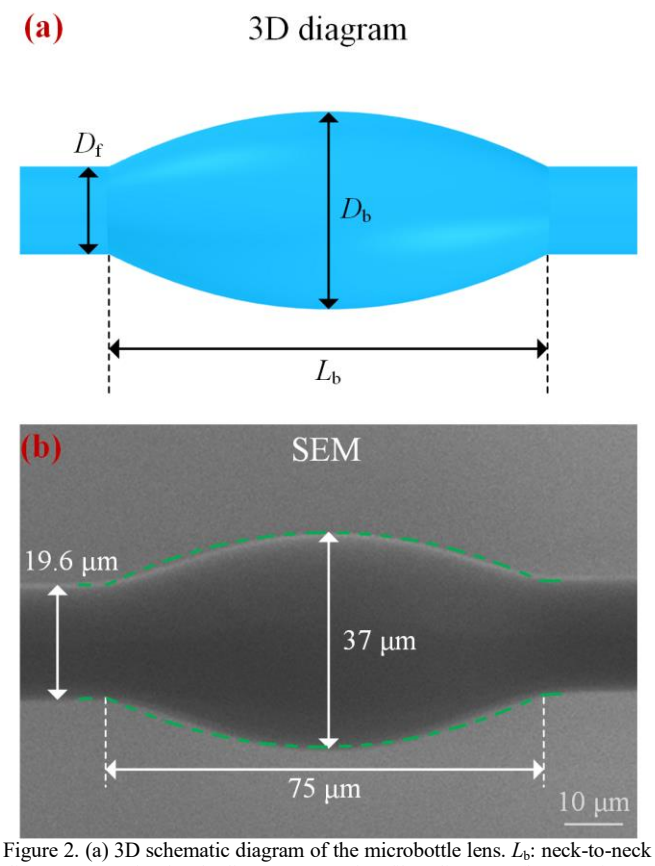

Figure 2. (a) 3D schematic diagram of the microbottle lens. L_b: neck-to-neck distance, D_b: microbottle diameter, D_f: microfiber diameter. (b) SEM image of a composite microbottle lens prepared by NOA 61 adhesive and silica microfiber with fitted parabolic profile (green dashed curve).

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III. Nano-imaging of Subwavelength Structures

A prepared microbottle lens with geometric dimensions of (L_b, D_b, D_f) = (56.7, 29.6, 14.1) μm, as shown in Fig. 3(a), is used to study the nano-imaging of subwavelength structures. One end of the microfiber is the microbottle lens, while the other end, as the stem, is fixed on a 3D nano-translation stage for precisely manipulating and positioning the microbottle lens. Two specimens used here are L-bar grating structures formed by transverse and longitudinal stripes perpendicular to each other. As illustrated in Figs. 3(b) and 3(c), the pitches of the grating structures are P = 360 nm and P = 260 nm, respectively. The microscope used is an upright Zeiss microscopy operating in reflection mode. A wide-field light with a central wavelength of 616.3 nm is focused through a 63× (NA = 1.0) objective lens and used to illuminate the microbottle lens and the specimen. The subwavelength features on the L-bar grating structure are captured by the microbottle lens and projected to form an enlarged image. The image process is observed by moving the microscope objective to the image plane and recorded with a high-performance microscopy camera.

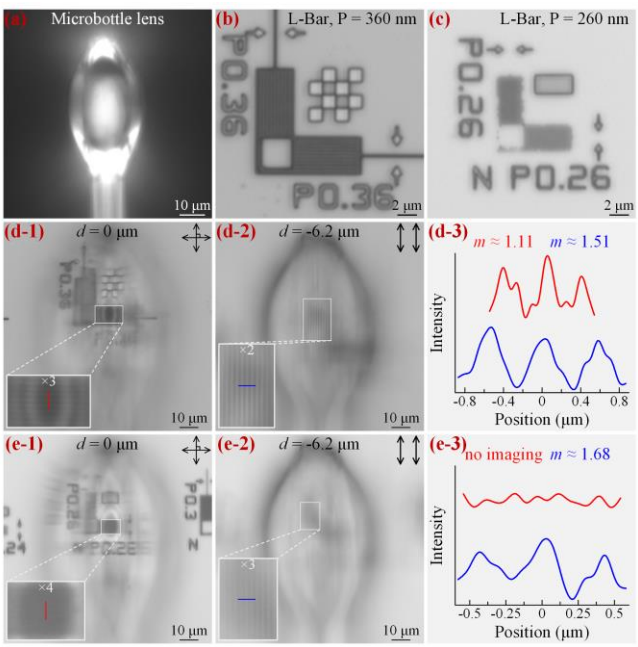

Fig. 3. (a) Microscopic image of a microbottle lens with (L_b, D_b, D_f) = (56.7, 29.6, 14.1) μm. (b) Microscopic images of the L-Bar grating structures with P = 360 nm. (c) Microscopic images of the L-Bar grating structures with P = 260 nm. (d-1) Images of the grating stripes of P = 360 nm at d = 0 μm. (d-2) Images of the grating stripes of P = 360 nm at d = -6.2 μm. (d-3) Irradiance intensity distributions along the red and blue solid lines in (d-1) and (d-2). (e-1) Images of the grating stripes of P = 260 nm at d = 0 μm. (e-2) Images of the grating stripes of P = 260 nm at d = -6.2 μm. The bottom left insets of (d-1), (d-2), (d-3) and (d-4) are the larger view of the annotated areas in each figure. (e-3) Irradiance intensity distributions along the red and blue solid lines in (e-1) and (e-2).

According to Abbe diffraction limit of λ/(2NA), the resolution of our microscope is ~313.15 nm. It can be seen from the microscopic images in Figs. 3(b) and 3(c) that the stripe structure with P = 360 nm can be resolved, while the stripe structure with P = 260 nm cannot be directly seen with the microscope. When the microbottle lens is utilized to image the grating stripes perpendicular [Fig. 3(d-1)] and parallel [Fig. 3(d-2)] to the microfiber axis, the corresponding images are observed at two different image plane positions of d = 0 and d = -6.2 μm. The symbol "d" represents the positions relative to specimen surface. As can be seen from the illustrations in Figs. 3(d-1) and 3(d-2), the sharpness and contrast of the image of the longitudinal stripes is higher than that of the transverse stripes. The corresponding magnifications, which are shown in Fig. 3(d-3), for the case of longitudinal stripes ($m \approx 1.51$) is higher than the case of transverse stripes ($m \approx 1.11$).

When the microbottle lens is used to image the subwavelength grating stripes with smaller pitch of P = 260 nm, the image of the stripes can be observed in the case of stripes parallel to the microfiber axis, but cannot be observed in the case of stripes perpendicular to the microfiber axis. As can be seen from the inset in Fig. 3(e-1), no image of the transverse stripes is obtained. The light intensity distribution of the red solid line plotted in Fig. 3(e-3) is correspondingly a small fluctuation curve. From the inset in Fig. 3(e-2), the grating strips are distinguished, and its intensity curves [the blue curve in Fig. 3(e-3)] clearly show three separate peaks. The image magnification is about 1.68.

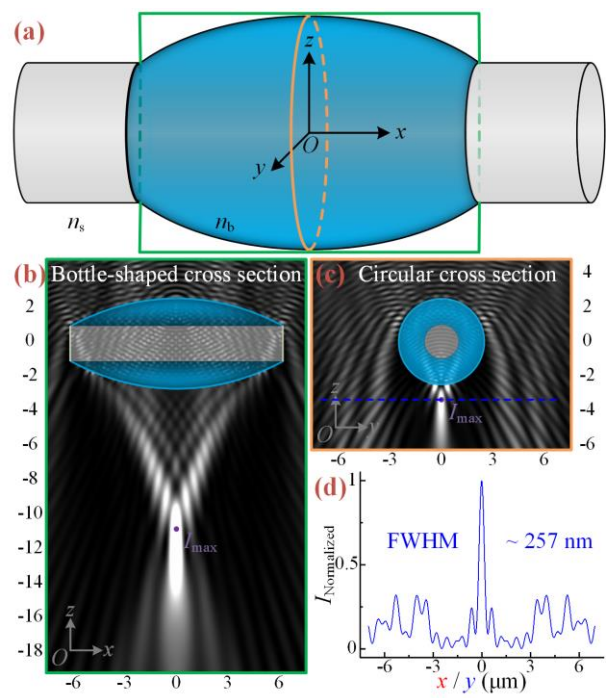

Fig. 4. (a) Schematic illustration of the cross-sections along and perpendicular to the microfiber axis of the microbottle lens. Light field distributions of (b) bottle-shaped cross section and (c) circular cross section of the microbottle lens obtained from FEM-based full wave simulation. (d) Normalized intensity profile along the dashed blue line in (c).

The better resolution of the microbottle lens in the direction perpendicular to the microfiber axis may be due to the higher curvature of the cross-section in the longitudinal direction compared with the transverse direction. Figure 4(a)

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is the schematic diagram of the cross-sections parallel and vertical to the microfiber axes, which are in the shapes of a bottle and a circle, respectively. We simulate the light field distributions when light passing by these two shapes by FEM. The light source used for the full-wave simulation is a monochromatic plane wave with wavelength λ = 616.3 nm and unit intensity. Figures 4(b) and 4(c) are the results of the simulation. It can be seen that the focal point and the full width at half-maximum (FWHM) of the microlens with a bottle profile are significantly larger than those with a circular profile. The FWHM value of the microlens with circular profile, as shown in Fig. 4(d), is ~ 257 nm, which is smaller than the pitch of L-Bar grating structures with P = 260 nm. Therefore, the longitudinal but not transverse stripes can be imaged with this microbottle lens.

IV. Conclusions

In this paper, we present a microbottle lens consisting of a silica core and a UV-curable polymer cladding layer, whose shape can be fitted with a parabolic profile. With this microbottle lens, the subwavelength grating stripes (P = 360 nm) with orientation perpendicular or parallel to the long axis of the microbottle lens, can be imaged at or below the surface of the sample. When the pitch of the grating stripes (P = 260 nm) is smaller than the diffraction limit (~313.15 nm), only the grating stripes parallel to the long axis can be imaged at the position below the sample surface. Through FEM-based full-wave simulation, the imaging capability of the microbottle lens in two imaging modes is analyzed. The microbottle lens, which enables two modes of imaging, has potential applications in areas such as optical encryption/decryption and anti-counterfeiting techniques combined with special structures.

Acknowledgment

This work was supported by National Natural Science Foundation of China (62175252, 62074155); Guangdong Basic and Applied Basic Research Foundation (2019A1515011242); Shenzhen Science and Technology Innovation Commission (KCXFZ202002011008124, JSGG20210713091807026, JCYJ20210324101405016).

References

[1] Z. Wang, W. Guo, L. Li, B. Luk'yanchuk, A. Khan, Z. Liu, Z. Chen and M. Hong, "Optical virtual imaging at 50 nm lateral resolution with a white-light nanoscope," Nat. Commun., vol. 2, pp. 218, 2011.

[2] Z. Wang, "Microsphere super-resolution imaging," Nanosci., vol. 3, pp. 193-210, 2016.

[3] B. S. Luk'yanchuk, R. Paniagua-Domínguez, I. Minin, O. Minin and Z. Wang, "Refractive index less than two: Photonic nanojets yesterday today and tomorrow," Opt. Mater. Express, vol. 7, no. 6, pp. 1820-1847, 2017.

[4] G. Huszka and M. A. M. Gijs, "Super-resolution optical imaging: A comparison," Micro Nano Eng. Vol. 2, pp. 7-28, 2019.

[5] L. Chen, Y. Zhou, Y. Li and M. Hong, "Microsphere enhanced optical imaging and patterning: From physics to applications," Appl. Phys. Rev., vol. 6, no. 2, pp. 021304, 2019.

[6] A. Darafsheh, "Microsphere-assisted microscopy," J. Appl. Phys., vol. 131, no. 3, pp. 031102, 2022.

[7] F. Wang, H. S. S. Lai, L. Liu, P. Li, H. Yu, Z. Liu, Y. Wang and W. J. Li, "Super-resolution endoscopy for real-time wide-field imaging," Opt. Express, vol. 23, no. 13, pp. 16803-16811, 2015.

[8] J. Monks, B. Yan, N. Hawkins, F. Vollrath and Z. Wang, "Spider silk: mother nature's bio-superlens", Nano Lett., vol. 16, no. 9, pp. 5842-5845, 2016.

[9] G. Gu, R. Zhou, H. Xu, G. Cai and Z. Cai, "Subsurface nano-imaging with self-assembled spherical cap optical nanoscopy," Opt. Express, vol. 24, no. 5, pp. 4937-4948, 2016.

[10] Y. Deng, S. Yang, Y. Xia, Y. Cao, J. Wang, F. Wang and Y.-H. Ye, "Super-resolution imaging properties of cascaded microsphere lenses," Appl. Opt., vol. 57, no. 20, pp. 5578-5582, 2018.

[11] Y. Li, X. Liu and B. Li, "Single-cell biomagnifier for optical nanoscopes and nanotweezers," Light Sci. Appl., vol. 8, no. 1, pp. 61, 2019.

[12] L. Li, W. Guo, Y. Yan, S. Lee and T. Wang, "Label-free super-resolution imaging of adenoviruses by submerged microsphere optical nanoscopy," Light Sci. Appl., vol. 2, no. 9, pp. e104, 2013.

[13] H. Yang, N. Moullan, J. Auwerx and M. A. M. Gijs, "Super-resolution biological microscopy using virtual imaging by a microsphere nanoscope," Small, vol. 10, no. 9, pp. 1712-1718, 2014.

[14] S. Lee and L. Li, "Rapid super-resolution imaging of sub-surface nanostructures beyond diffraction limit by high refractive index microsphere optical nanoscopy," Opt. Communi., vol. 334, pp. 253-257, 2015.

[15] W. Fan, B. Yan, Z. Wang and L. Wu, "Three-dimensional all-dielectric metamaterial solid immersion lens for subwavelength imaging at visible frequencies," Sci. Adv., vol. 2, no. 8, pp. e1600901, 2016.

[16] B. Hou, M. Xie, R. He, M. Ji, S. Trummer, R. H. Fink, and L. Zhang, "Microsphere assisted super-resolution optical imaging of plasmonic interaction between gold nanoparticles," Sci. Rep. vol. 7, pp. 13789 2017.

[17] S. Perrin, A. Leong-Hoi, S. Lecler and P. Montgomery, "Microsphere-assisted phase-shifting profilometry," Appl. Opt., vol. 56, no. 25, pp. 7249-7255, 2017.

[18] A. Leong-Hoi, C. Hairaye, S. Perrin, S. Lecler, P. Pfeiffer and P. Montgomery, "High resolution microsphere-assisted interference microscopy for 3D characterization of nanomaterials," Phys. Status Solidi A, vol. 215, no. 6, pp. 1700858, 2018.

[19] P. C. Montgomery, S. Lecler, A. Leong-Hoï and S. Perrin, "High resolution surface metrology using microsphere-assisted interference microscopy," Phys. Status Solidi A, vol. 216, no. 13, pp. 1800761, 2019.

[20] T. Zhang, H. Yu, P. Li, X. Wang, F. Wang, J. Shi, Z. Liu, P. Yu, W. Yang and Y. Wang, "Microsphere-based super-resolution imaging for visualized nano-manipulation," ACS Appl. Mater. Interfaces, vol. 12, no. 42, pp.48093-48100, 2020.

[21] S. Yang, Y.-H. Ye, J. Zang, Y. Pei, Y. Xia and J. Zhang, "Direct observation Brownian motion of individual nanoparticles in water using microsphere-assisted microscopy," Opt. Lett., vol. 46, no. 13, pp.3099-3102, 2021.

[22] S. Kwon, J. Park, K. Kim, Y. Cho and M. Lee, "Microsphere-assisted, nanospot, non-destructive metrology for semiconductor devices," Light Sci. Appl., vol. 11, no. 1, pp. 1-14, 2022.

[23] G. Gu, C. Guo, Z. Cai, H. Xu, L. Chen, H. Fu, K. Che, M. Hong, S. Sun and F. Li, "Fabrication of ultraviolet-curable adhesive bottle-like microresonators by wetting and photocuring," Appl. Opt., vol. 53, no. 32, pp. 7819-7824, 2014.

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